

NEXUS-1

Nuclear Plant Operator Console & Digital-Twin Platform

Project Roadmap

From Phase-0 demonstrator to an industrial-grade platform

Organised mode by mode · foundation first, then each operating mode adds its layer

.NET / C# (no Orleans) · Angular · MS SQL Server · Kafka · SignalR
OpenTelemetry → Prometheus → Grafana · Docker / Kubernetes

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ESTIMATE, NOT A QUOTE

Hours are planning estimates for senior engineers and shift with team, seniority and final scope. Scope is the industrial platform (monitoring / analytics / digital-twin / decision-support). The regulated, plant-connected deployment is a separate, specialist-led track and is not estimated in engineering hours here.

This edition reorganises the plan around the platform's six operating modes. A shared Foundation is built once; each mode then adds the capability that lights it up. A capability matrix shows exactly which screens and controls are present, read-only, or hidden in each mode — and why the platform can watch and advise but never control a real reactor.

How to read this roadmap

What changed, how the hours work, and what the plan delivers

This plan takes NEXUS-1 from its current state — a complete, single-file, client-side demonstrator (about 25 screens, the real point-kinetics engine, cross-module event behaviour, but no backend, persistence, auth, real-time transport, tests or infrastructure) — to an industrial-grade platform on the stated stack. Most of the work is not re-skinning tabs; it is building the distributed system the tabs sit on top of.

Reorganised mode by mode

The previous edition was phased (P0-P7). This edition keeps the same engineering but tells the story the way a reviewer actually reasons about the product: what can the platform do, and in what order does each capability become real? The answer is the six operating modes defined in the project specification. They are cumulative — each builds on the one before, and all of them stand on a common Foundation.

Foundation first, then each mode adds its layer

Shadow, Advisory, Supervisory and Hybrid all need the same backbone: identity and RBAC, the API gateway, the Kafka spine, the SignalR real-time path, the relational core on MS SQL Server, the telemetry historian, observability and the Angular shell. That backbone is built once, in the Foundation section, and is not re-charged to each mode. Every mode section below lists only the work specific to that mode, layered on top.

How the hours work

- Each task carries a Low and High estimate of focused engineering effort (design + build + test + review).
- The Planning figure per mode is the midpoint plus ~15% contingency — use it for budgeting.
- A separate line covers program management & technical leadership spread across all modes.
- Production Hardening & QA (load / soak / chaos / HA-DR / contract testing) is a cross-cutting track that scales with how many modes go live.
- Excluded from hours: hardware / licences, third-party services, and the regulatory track.

The locked technology stack

Layer	Choice
Back end	.NET / C# microservices (no Orleans); REST + gRPC; database-per-service ownership.
Front end	Angular single-page application; the existing design tokens ported into a shared library.
Real-time	SignalR fan-out for live push to the client (telemetry, presence, alarms).
Messaging	Kafka for everything — domain events, commands and the high-rate telemetry stream on one log (no Azure Service Bus).
Relational data	Microsoft SQL Server — configuration, master data, the hash-chained audit log, per-service schemas.
Telemetry historian	Time-series store for high-rate plant signals, fed from the Kafka stream (separate from ops metrics).
Observability	OpenTelemetry instrumentation → Prometheus metrics → Grafana dashboards & alerting.
Delivery	Docker + Kubernetes; per-service CI/CD; infrastructure-as-code.

The six operating modes

What each mode is, and the safety principle that runs through all of them

Mode	Status	What it is
Simulation	Active at Phase 0	The whole plant runs virtually. This is what NEXUS-1 is today — every value is generated in the browser; nothing touches a real reactor. Operators can drive the simulated reactor (rods, SCRAM) against the real point-kinetics model.
Training	Active at Phase 0	A teaching layer on the simulation. Five scored drills — Power Maneuver, Deep Power Reduction, Reactor Trip Response, Xenon Transient Management and Load-Follow — each with objectives, a live score and a debrief logged as a TRAINING audit entry.
Shadow	Roadmap	The platform mirrors a live plant's telemetry but cannot act on it — it only watches. All write affordances are removed, and the digital-twin reconciliation becomes meaningful: model expectation against real telemetry, where divergence is an early-warning signal.
Advisory	Roadmap	Shadow plus the analytics tier surfaced as a first-class operational layer: predictive maintenance, the dependency / causal model, the interpretable RL optimiser, and the retrieval-grounded DSLM assistant — all advisory. Humans make every decision.
Supervisory	Roadmap	Connected to control infrastructure for monitoring and coordination, still under human supervision and still with no control authority. This mode carries the full critical-infrastructure security posture (mTLS, segmentation, SIEM, penetration testing, OT policy).
Hybrid Digital Twin	Roadmap	Keeps the simulation continuously in step with live plant data — state estimation and data assimilation, with what-if branches off the live state and full-fidelity divergence warning.

No control authority — the principle behind the matrix. In every mode the software may watch and advise, but it never directly operates the reactor's protection systems. Those stay hardware-isolated, deterministic, certified and human-supervised. This is why the rod / SCRAM control row is hidden the moment the platform is pointed at a real plant — the safety stance is enforced by the interface itself, not merely documented.

Capability matrix

Which screens and controls are present, read-only, or hidden in each mode

The platform does not swap five separate menus in and out. It re-scopes one console: most screens persist across all modes — what changes is where their data comes from and whether their write controls are enabled. A single mode-gating layer drives this from one source of truth.

Screen / control	Sim	Train	Shadow	Advis.	Superv.	Hybrid
Overview / Plant Fleet	●	●	◐	◐	◐	●
Reactor & Kinetics (view)	●	●	◐	◐	◐	●
Rod / SCRAM control	●	●	○	○	○	●
Training drills	○	●	○	○	○	○
3D Reactor / Plant views	●	●	◐	◐	◐	●
Trends / Historian	◐	◐	◐	◐	◐	◐
Alarms & Events	●	●	◐	●	●	●
Compliance / Audit (append-only)	◐	◐	◐	◐	◐	◐
Asset · Health · Aging · Inspection	●	●	◐	◐	●	●
Personnel · Robotics · Zone Access	●	●	◐	◐	●	●
Incident & Root-Cause Graph	●	●	◐	●	●	●
Digital Twin (reconciliation)	◐	◐	◐	◐	◐	◐
AI · RL · Dependencies (advisory)	◐	◐	◐	●	●	●
Protocol gateways / OT security	○	○	◐	◐	●	◐

● Interactive — incl. simulated control / inputs ◐ Read-only — mirrors live data, no state change ○ Hidden — not available in this mode

Reading the matrix

- Rod / SCRAM is the decisive row. It is interactive only where the reactor is virtual (Sim, Train, and the simulation branch of Hybrid). Against a real plant — Shadow, Advisory, Supervisory — the control is genuinely absent, not greyed out. That gating is the no-control-authority principle, made visible.
- Going live hides write affordances; it does not delete screens. Alarms, Incident and the asset domains persist — they stop accepting simulated inputs and start mirroring real state.
- Some features are promoted, not added. The advisory tier (AI / RL / Dependencies) exists as a sandbox in Sim and is surfaced as a first-class operational layer from Advisory onward. The Digital Twin is illustrative in Sim and becomes the centrepiece in Shadow and Hybrid.
- One gating layer, not five menus. Each nav item and control declares the modes it is valid in (present / read-only / hidden); the shell renders accordingly. This keeps a single source of truth and avoids five navigation trees drifting out of sync.

Executive summary

Foundation built once; modes layered cumulatively on top

Section	Scope	Low	High	Planning
Foundation	Shared platform backbone — all live modes depend on it	890	1,495	1,370
Simulation Mode	The Phase-0 console productionised end-to-end	1,315	2,185	2,015
Training Mode	Scored-drill layer on the simulation	180	310	280
Shadow Mode	Read-only live mirror + protocol ingest	490	885	790
Advisory Mode	Analytics / RL / DSLM surfaced as advisory	490	845	770
Supervisory Mode	Coordination views + full OT security posture	420	740	665
Hybrid Twin Mode	Live-and-sim synchronisation	290	510	460
Production Hardening & QA	Cross-cutting; scales with live modes	510	900	810
Program management & technical leadership	All modes	520	820	770
TOTAL — industrial platform (all modes)		5,105	8,690	7,930

All figures in engineering work-hours. Planning = midpoint + ~15% contingency. Totals are higher than the previous phase-based edition because this plan makes genuinely new scope explicit — the Training drill engine, the Hybrid synchronisation mode, the DSLM assistant and a broader protocol set.

What is live at each step

Because the modes are cumulative, the platform reaches usable milestones long before the full total is spent:

When this is built...	...the platform can	Adds
Foundation + Simulation	A productionised single-plant console on real infrastructure — the Phase-0 app, but distributed, persisted, authenticated and streaming.	≈ 3,385 h
+ Training	Operators can run scored drills against the live model.	+ 280 h
+ Shadow	The console can mirror a real plant read-only; twin divergence becomes a live signal.	+ 790 h
+ Advisory	Predictive, RL and DSLM advice surfaced operationally — humans still decide.	+ 770 h
+ Supervisory	Coordinated integration under full critical-infrastructure security.	+ 665 h
+ Hybrid	Simulation and live data run synchronised; what-if off live state.	+ 460 h

› Build the Foundation + Simulation slice first. It converts every estimate below into measured velocity and yields a demonstrably industrial system before the live modes are attempted.

Architecture: microservices decomposition

Independently deployable services aligned to the menu's bounded contexts

Each service owns its data, publishes domain events to Kafka, and is reached synchronously only through the API gateway. Cross-cutting concerns (gateway, mesh, identity, observability) are shared platform, not per-service. Start coarse-grained — several services can begin merged and split later as load and team topology justify it.

Service	Responsibility	Backs (screens)
Identity & Access	AuthN/Z, RBAC, sessions, tokens	all (security)
API Gateway / BFF	Edge routing, aggregation, auth enforcement	all
Realtime / Notification	SignalR fan-out, presence, push	all live screens
Simulation / Digital Twin	Physics, point-kinetics, fleet state, twin	Overview, Reactor, Kinetics, 3D, Twin
Telemetry & Historian	Ingest, time-series store, queries	Trends, Power, Coolant, Steam
Command	Rod / SCRAM control path, validation	Control Rods (Sim / Hybrid only)
Alarm	Alarm rules, dedup, ack, escalation	Alarms & Events
Audit	Hash-chained tamper-evident log (SQL Server)	Compliance / Audit
Asset & Health	Component registry, live health, aging	Component Registry, Aging
Inspection	NDT records and inspection workflow	Rod Inspection
Personnel	Staffing, rosters, coverage rules	Personnel
Robotics	Robot / vehicle fleet, mission readiness	Robotics & Vehicles
Zone Access	Live presence, permissions matrix	Zone Access
Incident & Root-cause	Event store, timeline, causal graph	Incident, Root Cause Graph
Analytics / AI	Predictive maintenance, dependency / causal, DSLM	AI Diagnostics, Dependencies
RL Optimiser	Twin-only advisory optimisation sandbox	Optimization (RL)
Protocol Gateways	OPC-UA / MQTT / Modbus, read-only ingest	Shadow+ (integration)

> *Splitting too early is the most common microservices mistake and inflates the Foundation and QA lines. Begin with coarse-grained services; let load and team structure drive the splits.*

Detailed work breakdown

Foundation first, then each operating mode — Low / High hours per task

F Foundation — shared platform backbone

Built once. Every live mode depends on it; not re-charged per mode.

ID	Task	Low	High	Notes
F1	Architecture, ADRs, service decomposition, golden-path scaffold, standards	70	120	Boundaries, ownership, topic design
F2	Dev env + CI/CD + IaC + K8s baseline + gateway / discovery / config	95	160	docker-compose, per-service pipelines
F3	Identity / Auth + RBAC model (baseline)	45	80	Keycloak / Duende / Entra
F4	Event & telemetry contracts + schema registry	30	50	Avro / Protobuf, versioning
F5	Kafka spine — events + commands + telemetry, registry, idempotency, outbox	90	140	One log for all messaging
F6	SignalR hubs + backpressure + throughput spike	60	100	Prove the real-time spine early
F7	Relational core on MS SQL Server — domain model, persistence, migrations	70	115	DB-per-service, no shared schema
F8	Telemetry historian write path + retention / downsampling	50	85	Continuous aggregates
F9	Read models, caching, health checks, API error model, config, feature flags	55	95	Query side + cross-cutting
F10	Inter-service comms + consistency (outbox / saga, sync gRPC / REST)	50	85	Eventual consistency
F11	Observability — OpenTelemetry → Prometheus → Grafana, dashboards, alerting	60	100	Ops metrics, not plant telemetry
F12	Service mesh + resilience patterns (circuit breakers, retries, bulkheads)	45	80	
F13	Angular shell — routing / menu mirror, design system, OIDC, guards, API + SignalR client, state store	140	230	Port existing CSS tokens
F14	Mode / capability gating engine (present / read-only / hidden)	30	55	Drives the capability matrix
Mode / layer subtotal		890	1,495	Planning ≈ 1,370 h

› Highest leverage of the whole programme — a weak Foundation multiplies cost in every mode.

› Output: a running, authenticated, observable skeleton that streams over SignalR and Kafka.

M1 Simulation Mode — the Phase-0 console, productionised

Active today as a demonstrator; here it becomes a distributed, persisted, tested system.

ID	Task	Low	High	Notes
S1	Simulation / Twin service — fleet model & master tick	70	110	Port the multi-unit simulation
S2	Point-kinetics engine port + unit tests	90	140	HIGH RISK — 6 delayed groups, feedbacks
S3	Thermal-hydraulics, secondary & grid models	50	80	Coolant / steam / power
S4	Deterministic sim test harness (golden-master)	40	70	Reproducibility & regression
S5	Command path (rod / SCRAM) + audit emission	40	70	Interactive only in Sim / Train / Hybrid-sim
S6	Screens: Overview / Fleet / Power	60	100	
S7	Screens: Core / Rods / Neutronics / Steam / Coolant	80	130	Interactive controls
S8	Kinetics + Model Analysis screens	55	95	Inhour check, sensitivity, margins
S9	3D screens (Reactor 3D + Plant 3D)	90	150	three.js, performance
S10	Trends / History (time-series)	40	70	
S11	Alarm engine + ack / escalation + UI	90	140	Dedup, severity, routing
S12	Tamper-evident audit (hash-chained) + UI	70	110	Real crypto + verification
S13	Asset / component registry + live health + UI	70	110	Per-unit equipment
S14	Aging & degradation service + UI	50	80	
S15	Inspection / NDT records + workflow + UI	60	100	
S16	Personnel / staffing + coverage + UI	60	100	
S17	Robotics & vehicles + readiness + UI	60	100	
S18	Zone access (presence + permissions) + UI	50	90	
S19	Incident analysis (event store + timeline) + UI	50	90	Live SCRAM events
S20	Root-cause graph service + ranking + UI	70	120	Fault-tree backbone, causal-origin ranking
S21	Frontend tests (unit / component / e2e) + i18n (EN / EL)	70	130	Cypress / Playwright
Mode / layer subtotal		1,315	2,185	Planning ≈ 2,015 h

› The point-kinetics port (S2) is the single biggest technical risk — protect its schedule, test it hard.

› Output: the entire current console, production-grade, on real data — the cross-module SCRAM ripple intact.

M2 Training Mode — scored-drill layer

A teaching layer on the simulation; small, because the hard parts already exist in Sim.

ID	Task	Low	High	Notes
T1	Scenario / drill engine + state hooks	50	85	Inject conditions into the live model
T2	Five scored drills (objectives, scoring, live evaluation)	60	100	Power maneuver ... load-follow
T3	Debrief + TRAINING audit entries + replay	30	55	
T4	Instructor controls + scenario authoring UI	40	70	
Mode / layer subtotal		180	310	Planning $\approx$ 280 h

› Depends only on Simulation Mode — no new infrastructure.

› Output: operators fly the reactor against scored objectives, each run logged for review.

M3 Shadow Mode — read-only live mirror

The platform watches a real plant and cannot act on it. The twin's divergence becomes real.

ID	Task	Low	High	Notes
SH1	Protocol gateway architecture + isolation	40	70	Isolated, read-only by construction
SH2	OPC-UA gateway (read-only)	80	140	
SH3	MQTT gateway (read-only)	50	90	
SH4	Modbus TCP gateway (read-only)	50	90	
SH5	IEC 61850 / DNP3 adapters (extensible)	60	110	Optional per site
SH6	Normalisation / validation to canonical model	60	110	
SH7	Read-only twin sync to live data (divergence / residuals real)	70	130	Early-warning signal
SH8	Write-affordance suppression + read-only enforcement tests	30	55	Enforces no-control-authority
SH9	Conformance / integration test rig	50	90	
Mode / layer subtotal		490	885	Planning $\approx$ 790 h

› Strictly read-only; there is no command path to plant equipment.

› Output: the Digital Twin reconciles model against real telemetry; all write controls disappear.

M4 Advisory Mode — analytics surfaced, humans decide

The analytics tier promoted from sandbox to a first-class operational layer.

ID	Task	Low	High	Notes
A1	Feature pipeline off the historian	60	110	Time-series → features
A2	Predictive-maintenance baselines + model serving	90	150	Statistical first, then ML
A3	Dependency / causal service (graph + weights)	60	100	
A4	RL optimiser (twin-only, advisory, readable policy)	90	150	Interpretable Q-table, safety-clamped
A5	Model registry, evaluation, drift monitoring	50	90	
A6	DSL assistant (Ollama, air-gapped, retrieval-grounded, cited)	90	160	Production target; cites registry / incidents / docs
A7	Advisory UI surfacing + guardrail labelling + acceptance audit	50	85	Every accepted advisory logged
Mode / layer subtotal		490	845	Planning ≈ 770 h

› Realism is bound by data — start transparent / heuristic, become ML once historian data exists.

› Advisory only: never wired to control or safety actuation.

M5 Supervisory Mode — coordination + full OT security

Connected for monitoring and coordination, still human-supervised, still no control authority.

ID	Task	Low	High	Notes
SU1	Supervisory coordination & orchestration views	60	110	
SU2	Gateway connection management / health (elevated)	40	70	
SU3	mTLS / service identity, secrets (KMS), HSM	70	120	
SU4	Segmentation / zero-trust, gateway hardening	50	90	
SU5	Security audit coverage + SIEM hooks	40	70	
SU6	Threat model + SCA / SAST / DAST in CI	50	90	
SU7	External penetration test + remediation	80	140	Third-party engagement
SU8	OT security policy pack + documentation	30	50	
Mode / layer subtotal		420	740	Planning ≈ 665 h

› Carries the critical-infrastructure security posture; the pen test is best near feature-complete.

› Output: coordinated integration with control infrastructure — monitoring and orchestration, not control.

M6 Hybrid Digital Twin Mode — live and sim synchronised

Keeps the simulation continuously in step with live plant data.

ID	Task	Low	High	Notes
H1	State estimation / data assimilation (sim kept in step with live)	90	150	Filtering / nudging
H2	Continuous reconciliation + bias / drift correction	60	110	
H3	What-if branching off live state	50	90	Predictive scenarios from now
H4	Divergence-driven early-warning at full fidelity + UI	50	90	
H5	Hybrid-mode validation harness	40	70	
Mode / layer subtotal		290	510	Planning ≈ 460 h

› The richest mode — and the one most dependent on everything before it.

› Output: a twin that tracks the plant and can branch into forecasts off the live state.

Q Cross-cutting — Production Hardening & QA

Scales with how many modes go live; runs largely in parallel with the mode work.

ID	Task	Low	High	Notes
Q1	Test strategy + pyramid (unit / integration / contract)	90	150	
Q2	Load & soak testing (Kafka / SignalR fan-out)	70	120	
Q3	Chaos / resilience + failover drills	50	90	
Q4	HA / redundancy + backpressure tuning	60	100	
Q5	Kubernetes production deploy (Helm, autoscaling)	80	140	
Q6	DR + backup / restore + air-gapped support	60	110	
Q7	Documentation, runbooks, ops, API reference	60	110	
Q8	Consumer-driven contract testing (Pact)	40	80	Cross-service contracts in CI
Mode / layer subtotal		510	900	Planning ≈ 810 h

› Output: a deployable, observable, resilient platform across whichever modes are enabled.

PM Cross-cutting — Program management & technical leadership

Spread across all modes.

ID	Task	Low	High	Notes
PM1	Architecture governance, ADR review, cross-team coordination	220	350	
PM2	Planning, tracking, risk & dependency management	180	280	
PM3	Stakeholder comms, demos, documentation oversight	120	190	
Mode / layer subtotal		520	820	Planning ≈ 770 h

Separate track — Nuclear V&V / regulated deployment

A different undertaking from the engineering above

Connecting the platform to, and having it trusted by, a real nuclear plant is largely organisational and specialist-led. It is not estimated here in engineering hours, because doing so honestly requires a nuclear QA organisation rather than a software team. Indicative activities:

- Formal verification & validation of the simulation / twin models against qualified references.
- Requirements traceability and a documented safety / security case.
- Alignment with the relevant standards (e.g. IEC 61513 system level, IEC 60880 software, IEC 62645 cyber-security).
- Independent verification and assessment by a separate competent party.
- Regulator engagement and licensing-basis documentation.
- Even then, the platform remains in the monitoring / analytics / supervisory layer; certified reactor-protection functions stay hardware-isolated, deterministic and outside this system.

Indicative duration: 12–36+ months, specialist-led, partly parallel with later engineering. Treat as a programme, not a sprint.

Assumptions, caveats & authorship

- These are planning estimates, not a fixed-price quotation. Actuals depend on team seniority, the final agreed scope, and the definition of 'complete'.
- The point-kinetics port (S2) is the dominant technical risk; the alarm engine and cryptographic audit (S11, S12) and the real-time pipeline (F5, F6, Q2) are the next most uncertain.
- Kafka carries all messaging — events, commands and the high-rate telemetry stream — on one log. With a single simulation producer this is future-proofing for multi-source ingestion, not a present functional need; it adds operational weight early but suits the architecture and integration goals.
- Microservices add platform, contract-testing and coordination overhead. Begin coarse-grained and split only when load or team topology demands it; over-decomposition is the main way this estimate would overrun.
- Re-validate every estimate at the end of the Foundation + Simulation slice — that slice converts these ranges into measured velocity.

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